IIR design: bilinear transform and Butterworth

Answers and worked solutions · Day 19 · Wed 11 Nov · pack D11

5. Practice

From your course files

T9 Q4 (tutorial).

Hint: only the frequencies are warped. The magnitude limits stay the same.

Bilinear transformation requires frequency prewarping, $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$:

$$0.8 \le |H(\Omega)| \le 1, \quad 0 \le \Omega \le \frac{2}{T}\tan 0.15\pi \;\; (= 1.0191 \text{ with } T = 1)$$

$$|H(\Omega)| \le 0.1, \quad \Omega \ge \frac{2}{T}\tan 0.3\pi \;\; (= 2.7528 \text{ with } T = 1).$$

T9 Q2 (tutorial).

Hint: the official solution uses $2/T = 20$ (that is, $T = 0.1$). Map the quadratic with $b = 0.2$, $c = 9.01$.

Substitute $s = 20\,\dfrac{1 - z^{-1}}{1 + z^{-1}}$ and multiply top and bottom by $(1 + z^{-1})^2$:

- numerator: $\left[20(1 - z^{-1}) + 0.1(1 + z^{-1})\right](1 + z^{-1}) = 20.1 + 0.2z^{-1} - 19.9z^{-2}$
- denominator: $400(1 - z^{-1})^2 + 0.2 \cdot 20(1 - z^{-2}) + 9.01(1 + z^{-1})^2 = 413.01 - 781.98z^{-1} + 40$

$$H(z) = \frac{20.1 + 0.2z^{-1} - 19.9z^{-2}}{413.01 - 781.98z^{-1} + 405.01z^{-2}} = \frac{0.0487 + 0.0005z^{-1} - 0.0482z^{-2}}{1 - 1.8934z^{-1} + 0.9806z^{-2}} \quad (\text{after normalising } a_0 = 1)$$

Zeros $z_1 = -1$, $z_2 = 0.99$; poles $p_{1,2} = 0.9467 \pm j0.2905$.

T9 Q3 (tutorial).

Hint: warping makes the digital integrator fall to zero at $\omega = \pi$. Avoid inputs with DC (pole at $z = 1$).

(a)

$$H(z) = H_a(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{T}{2}\,\frac{1 + z^{-1}}{1 - z^{-1}}, \qquad |z| > 1 \text{ (causal system)}$$

$$y(n) - y(n-1) = \frac{T}{2}x(n) + \frac{T}{2}x(n-1).$$

(b) $H_a(\Omega) = \dfrac{1}{j\Omega} = -\dfrac{j}{\Omega}$, so $|H_a(\Omega)| = \dfrac{1}{|\Omega|}$ (infinite at $\Omega = 0$, a hyperbola either side) and $\theta(\Omega) = -\frac{\pi}{2}$ for $\Omega > 0$, $+\frac{\pi}{2}$ for $\Omega < 0$.

(c)

$$H(\omega) = \frac{T}{2}\,\frac{1+e^{-j\omega}}{1-e^{-j\omega}} = \frac{T}{2}\,\frac{e^{-j\omega/2}\left(e^{j\omega/2}+e^{-j\omega/2}\right)}{e^{-j\omega/2}\left(e^{j\omega/2}-e^{-j\omega/2}\right)} = -j\,\frac{T}{2}\,\frac{\cos(\omega/2)}{\sin(\omega/2)} = \frac{T}{2}\cot(\omega/2)\,e^{-j\pi/2}$$

$$|H(\omega)| = \frac{T}{2}|\cot(\omega/2)|, \quad -\pi \le \omega \le \pi; \qquad \theta(\omega) = \begin{cases} -\frac{\pi}{2}, & 0 < \omega \le \pi \\ \frac{\pi}{2}, & -\pi \le \omega < 0 \end{cases}$$

Strictly, $H(\omega)$ does not exist because the ROC $|z| > 1$ excludes the unit circle, but it can still be computed if the pole at $z = 1$ is avoided (L04 slides 12–13).

(d) For $T = 1$ ($\omega = \Omega$), $1/|\Omega|$ and $\frac{T}{2}|\cot(\omega/2)|$ match very well at all frequencies except near $\omega = \pi$, where the frequency warping of the bilinear transformation makes the digital integrator decay to zero. The phase responses are identical.

(e) Input signals with a non-zero DC component should be avoided, because they would make the digital integrator unstable (unbounded output).

T9 Q6 (tutorial).

Hint: radius $= \Omega_c$. 100 Hz $= 200\pi$ rad/s.

$\Omega_c = 50$ rad/s. 80 dB at 100 Hz means $10\log_{10}|H(200\pi)|^2 = -80$ dB:

$$\frac{1}{1+\left(\frac{200\pi}{50}\right)^{2N}} = 10^{-8} \quad \Rightarrow \quad N = \frac{\log(10^8-1)}{2\log(4\pi)} = 3.6390.$$

Rounding up, $N = 4$.

T9 Q5 (tutorial).

Hint: factor method: $(s + 1.2799)$ and $(s^2 + 1.2799s + 1.6380)$ with $K = 2$. Denominator constant 26.8876.

Edges. $\omega_p = \dfrac{2\pi\cdot 3}{30} = \dfrac{\pi}{5}$, $\omega_s = \dfrac{2\pi\cdot 9}{30} = \dfrac{3\pi}{5}$. Let $T = 1$ and prewarp:
$\Omega_p = 2\tan\frac{\pi}{10} = 0.6498$, $\Omega_s = 2\tan\frac{3\pi}{10} = 2.7528$.

Linear scale. $\delta_p = 10^{-1/20} = 0.8913$, $\delta_s = 10^{-20/20} = 0.1$.

Order.

$$N = \frac{\log\left[\dfrac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)}\right]}{2\log\dfrac{\Omega_s}{\Omega_p}} = 2.0596 \;\Rightarrow\; N = 3.$$

Cutoff. For the bilinear transformation it is fine to exceed the passband spec (no aliasing), so meet the stopband exactly:

$$\Omega_c = \Omega_s\left(\frac{\delta_s^2}{1-\delta_s^2}\right)^{1/(2N)} = 1.2799.$$

Analogue filter. Poles at -1.2799 and $1.2799e^{\pm j2\pi/3}$:

$$H(s) = \frac{1.2799^3}{(s+1.2799)(s^2+1.2799s+1.6380)} = \frac{2.0966}{s^3+2.5598s^2+3.2762s+2.0966}.$$

Bilinear ($s = 2\frac{1-z^{-1}}{1+z^{-1}}$, multiply through by $(1+z^{-1})^3$):

$$H(z) = \frac{2.0965+6.2895z^{-1}+6.2895z^{-2}+2.0965z^{-3}}{26.8876-21.3974z^{-1}+13.4983z^{-2}-2.2167z^{-3}} = \frac{0.0780+0.2339z^{-1}+0.2339z^{-2}+0.0780z^{-3}}{1-0.7958z^{-1}+0.5020z^{-2}-0.0824z^{-3}}$$

Difference equation.

$$y(n) - 0.7958y(n-1) + 0.5020y(n-2) - 0.0824y(n-3) = 0.0780x(n) + 0.2339x(n-1) + 0.2339x(n-2) + 0.0780x(n-3)$$

New questions

Question 1

a. Prewarp: $\Omega_p = 2\tan(0.1\pi) = 0.64984$ and $\Omega_s = 2\tan(0.25\pi) = 2.00000$.

$$0.85 \le |H(\Omega)| \le 1,\;\; 0 \le \Omega \le 0.6498; \qquad |H(\Omega)| \le 0.15,\;\; \Omega \ge 2.0000$$

(In dB: passband ripple 1.41 dB, stopband attenuation 16.48 dB.)

b. $\dfrac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)} = \dfrac{0.7225\times 0.9775}{0.0225\times 0.2775} = 113.11$, and $\Omega_s/\Omega_p = 3.07768$.

$$N = \frac{\log 113.11}{2\log 3.07768} = 2.1030 \Rightarrow \mathbf{N = 3}$$

Passband exact: $\Omega_c = 0.64984\left(\frac{0.7225}{0.2775}\right)^{1/6} = 0.7622$. Stopband exact: $\Omega_c = 2\left(\frac{0.0225}{0.9775}\right)^{1/6} = 1.0667$.

Any $0.7622 \le \Omega_c \le 1.0667$ meets both specs. (scipy `buttord` gives $N = 3$.)

Question 2

a. $\omega_c = 2\pi(1)/8 = \pi/4$. So $\Omega_c = \frac{2}{T}\tan(\omega_c/2) = 16000\tan(\pi/8) =$ **6627.4 rad/s**. (This is 1054.8 Hz, not 1000 Hz. That is warping at work.)

b. $K = 2/T = 16000$. The first-order factor maps to $(K + \Omega_c) + (\Omega_c - K)z^{-1} = 22627.4 - 9372.6z^{-1}$. The numerator becomes $\Omega_c(1 + z^{-1}) = 6627.4(1 + z^{-1})$.

$$H(z) = \frac{0.29289(1 + z^{-1})}{1 - 0.41421z^{-1}}, \qquad y(n) = 0.41421\,y(n-1) + 0.29289\,x(n) + 0.29289\,x(n-1)$$

Checks: $H(1) = 0.58579/0.58579 = 1$ (DC gain 1). $H(-1) = 0$ (zero at $\omega = \pi$). $|H(e^{j\pi/4})| = -3.01$ dB. The pole at $z = 0.414$ is stable.

c. The digital cutoff is $\omega = 2\arctan(\Omega_c T/2) = 2\arctan(\pi/8) = 0.74839$ rad, so $f = 0.74839 \times 8000/(2\pi) =$ **952.9 Hz**. That is 47 Hz low.

Question 3

a. $\Omega_c = 3141.59$ rad/s. The angles are $\frac{\pi}{2} + \frac{(2k+1)\pi}{8}$, that is $112.5^\circ, 157.5^\circ, 202.5^\circ, 247.5^\circ$:

$$s_{0,3} = -1202.2 \pm j2902.5, \qquad s_{1,2} = -2902.5 \pm j1202.2 \text{ rad/s}$$

b. Each pair gives $s^2 - 2\Omega_c\cos\theta\, s + \Omega_c^2$, with $\Omega_c^2 = 9.8696 \times 10^6$:

$$H(s) = \frac{9.7409 \times 10^{13}}{(s^2 + 2404.5s + 9.8696 \times 10^6)(s^2 + 5804.9s + 9.8696 \times 10^6)}$$

(The numerator is Ω_c^4.)

c. $\Omega/\Omega_c = 1500/500 = 3$, so the attenuation is $10\log_{10}(1 + 3^8) =$ **38.17 dB**.

d. $N = \log(10^6 - 1)/(2\log 3) = 6.2877 \Rightarrow N = 7$. (With $N = 4$, 60 dB is only reached at 2812 Hz.)

Question 4

a. $\omega_p = 2\pi/8 = \pi/4$ and $\omega_s = 6\pi/8 = 3\pi/4$. $\Omega_p = 2\tan(\pi/8) = 0.82843$ and $\Omega_s = 2\tan(3\pi/8) = 4.82843$. The ratio is 5.82843.

$$N = \frac{\log(99/0.25893)}{2\log 5.82843} = 1.6867 \Rightarrow \mathbf{N = 2}$$

b. Stopband exact: $\Omega_c = 4.82843/99^{1/4} = \mathbf{1.53072}$. So $\Omega_c^2 = 2.34312$ and $\sqrt{2}\Omega_c = 2.16477$. The poles are $-1.08239 \pm j1.08239$.

$$H(s) = \frac{2.34312}{s^2 + 2.16477s + 2.34312}$$

c. With $K = 2$: $4 + 4.32954 + 2.34312 = 10.67266$; $2(2.34312 - 4) = -3.31377$; $4 - 4.32954 + 2.34312 = 2.01358$.

$$H(z) = \frac{2.34312(1 + z^{-1})^2}{10.67266 - 3.31377z^{-1} + 2.01358z^{-2}} = \frac{0.21954 + 0.43909z^{-1} + 0.21954z^{-2}}{1 - 0.31049z^{-1} + 0.18867z^{-2}}$$

$$\boxed{y(n) = 0.31049\,y(n-1) - 0.18867\,y(n-2) + 0.21954\,x(n) + 0.43909\,x(n-1) + 0.21954\,x(n-2)}$$

d. At Ω_p: $-10\log_{10}[1 + (0.82843/1.53072)^4] = -0.357$ dB (better than the 1 dB allowed). At Ω_s: $-10\log_{10}[1 + (4.82843/1.53072)^4] = -20.00$ dB (met exactly). The poles are $0.1552 \pm j0.4057$, with $|z| = 0.434 < 1$, so the filter is stable.

VERIFIED with scipy: `butter(2, 1.53072, analog=True)` followed by `bilinear(..., fs=1)` gives the same coefficients, and `freqz` gives −0.357 dB and −20.00 dB at $\pi/4$ and $3\pi/4$.

Question 5

a. $\omega_p = 2\pi(2)/12 = \pi/3$ and $\omega_s = 2\pi(4)/12 = 2\pi/3$. $\Omega_p = 2\tan(\pi/6) = 1.15470$ and $\Omega_s = 2\tan(\pi/3) = 3.46410$. The ratio is exactly 3.

$$N = \frac{\log(99/0.25893)}{2\log 3} = 2.7063 \Rightarrow \mathbf{N = 3}$$

b. Passband exact: $\Omega_c = 1.15470/(0.25893)^{1/6} = \mathbf{1.44635}$. The LHP poles are at $120^\circ, 180^\circ, 240^\circ$: $-0.72318 \pm j1.25258$ and -1.44635. With $\Omega_c^2 = 2.09193$ and $\Omega_c^3 = 3.02566$:

$$H(s) = \frac{3.02566}{(s + 1.44635)(s^2 + 1.44635s + 2.09193)} = \frac{3.02566}{s^3 + 2.89270s^2 + 4.18386s + 3.02566}$$

c. Map the factors with $K = 2$:

- linear: $(2 + 1.44635) + (1.44635 - 2)z^{-1} = 3.44635 - 0.55365z^{-1}$
- quadratic: $4 + 2.89270 + 2.09193 = 8.98463$; $2(2.09193 - 4) = -3.81614$; $4 - 2.89270 + 2.09193 = 3.19923$

Multiply the two:
$(3.44635 - 0.55365z^{-1})(8.98463 - 3.81614z^{-1} + 3.19923z^{-2}) = 30.96419 - 18.1260$
.

The numerator is $3.02566(1 + 3z^{-1} + 3z^{-2} + z^{-3})$. Divide by 30.96419:

$$H(z) = \frac{0.097715 + 0.293145z^{-1} + 0.293145z^{-2} + 0.097715z^{-3}}{1 - 0.585389z^{-1} + 0.424312z^{-2} - 0.057203z^{-3}}$$

$$\begin{aligned} y(n) = {} & 0.585389\,y(n-1) - 0.424312\,y(n-2) + 0.057203\,y(n-3) \\ & + 0.097715\,x(n) + 0.293145\,x(n-1) + 0.293145\,x(n-2) + 0.097715\,x(n-3) \end{aligned}$$

d. At Ω_p: $-10\log_{10}[1 + (1.15470/1.44635)^6] = -1.00$ dB (met exactly). At Ω_s: $-10\log_{10}[1 + (3.46410/1.44635)^6] = -22.78$ dB (meets ≤ -20 dB). The poles in z are 0.16065 (from the linear factor, $0.55365/3.44635$) and $0.21237 \pm j0.55765$ ($|z| = 0.597$). All are inside the unit circle, so the filter is stable. DC gain: $\sum b_k = 0.78172 = \sum a_k$, so the gain is 1.

e. With $T = 1/12000$: $\Omega_p = 24000\tan(\pi/6)$ and $\Omega_s = 24000\tan(\pi/3)$. Both scale by 12000, so the ratio and N are unchanged, and $\Omega_c = 17356$ rad/s (also 12000 times larger). In the map, $K = 24000$ also scales by 12000. Every factor coefficient therefore scales by the same power of 12000, which cancels on normalising. scipy gives identical coefficients. The 3-dB point is at $\omega = 2\arctan(\Omega_c T/2) = 2\arctan(0.72318) = 0.3986\pi$, so $f = 0.3986 \times 6000 =$ **2392 Hz**.

> **VERIFIED:** `buttord(1/3, 2/3, 1, 20)` gives $(3,\ 0.39859)$, and `butter(3, 0.39859)` matches the coefficients above.

Question 6

a. The ratio is $\Omega_s/\Omega_p = 2$, so $N = \log(99/0.25893)/(2\log 2) = 4.289 \Rightarrow N = 5$. That is two orders more than necessary, because without prewarping the transition band looks narrower than it really is.

b. $\Omega_c = (\pi/3)/(0.25893)^{1/6} = 1.31169$. After the bilinear map, the digital gain at ω equals the analogue gain at $2\tan(\omega/2)$, **not** at ω:

- at ω_p: $\Omega = 1.15470$, so the gain is $-10\log_{10}[1 + (1.15470/1.31169)^6] =$ **−1.66 dB**. **This fails the 1 dB ripple spec.**
- at ω_s: $\Omega = 3.46410$, so the gain is $-10\log_{10}[1 + (3.46410/1.31169)^6] =$ **−25.32 dB** (passes).

The digital 3-dB point moved down to $2\arctan(1.31169/2) = 0.3695\pi$, compared with 0.3986π in the correct design. Warping compresses the frequency axis, so a filter designed with unwarped edges cuts off too early. (Checked with `freqz`: −1.660 dB and −25.318 dB.)

6. Quick recall check

Question

1. $s = 0 \to z = 1$ (DC gain kept). $s = \infty \to z = -1$ (zeros at $\omega = \pi$).
2. $\Omega = \frac{2}{T}\tan(\omega/2)$. It changes only the band-edge frequencies. δ_p and δ_s are unchanged.
3. $\delta_p = 10^{-1/20} = 0.8913$, and $(1 - \delta_p^2)/\delta_p^2 = 10^{0.1} - 1 = 0.2589$.
4. A circle of radius Ω_c centred on the origin, in the LHP. For $N = 3$ the angles are 120°, 180° and 240°.
5. $N = 4$. You always round up, because $N = 3$ would miss the spec.